



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

March 11, 2024

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 8220
(THIRTEENTH REVISION)

EXPIRATION DATE: 2028-02-29

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Applied Pressure Vessels, Inc
Valencia, CA
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, marking, sale and use of a non-DOT specification cylinder conforming with all regulations applicable to a DOT specification 3HT cylinder, except as specified herein, for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a) party status may not be granted to a manufacturing permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.302a in that a non-DOT specification cylinder is not authorized, except as specified herein.

5. BASIS: This special permit is based on the application of Applied Pressure Vessels, Inc dated January 4, 2024, and submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Air, compressed	2.2	UN1002	n/a
Nitrogen, compressed	2.2	UN1066	n/a

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Packagings prescribed is a non-DOT specification girth welded steel cylinder having a maximum water capacity of 25 cubic inches made in accordance with Applied Equipment Corporation's design specification including Drawing 54080-125-1 Rev. 0 dated September 11, 1974; and a non-DOT specification girth welded steel cylinder having a maximum water capacity of 20 cubic inches made in accordance with Allied Companies Drawing 54080-365-14 Rev. A dated August 30, 1990. Cylinder drawings are on file with the Office of Hazardous Materials Safety (OHMS) and cylinders are in full compliance with DOT Specification 3HT (§ 178.35 and § 178.44), except as follows:

§ 178.35(e) *Pressure relief devices*

(Add) A statement verifying the adequacy of the pressure relief devices for each cylinder must be submitted to the Associate Administrator for Hazardous Materials Safety (AAHMS) prior to initial shipment.

§ 178.44(a) *Type, size and service pressure.*

Cylinder must be girth welded and have one end fitting welded either as shown in:

(1) Drawing 54080-125-1 Rev. O with a maximum water capacity of 25 cubic inches; or

(2) Drawing 54080-365-14 Rev. A with a maximum water capacity of 20 cubic inches.

Service pressure may not exceed 6,000 psi.

§ 178.44(b) *Authorized steel*.

Austenitic stainless steel (21-6-9) with the following composition and properties is authorized.

<u>CHEMICAL ANALYSIS</u>	<u>PERCENT</u>
Carbon	0.040 max.
Manganese	8.00/10.00
Phosphorus	0.060 max.
Sulfur	0.030 max.
Silicon	1.00 max.
Chromium	19.00/21.50
Nickel	5.50/7.50
Nitrogen	0.15/0.40

TYPICAL MECHANICAL PROPERTIES

<u>Condition</u>	<u>Ultimate Tensile (PSI)</u>	<u>Yield Strength (PSI)</u>	<u>Elongation Percent In 2 in.</u>
Annealed	112,000	68,000	44
20% Cold- Reduction	150,000	120,000	18

§ 178.44(d) *Manufacture*.

(Add) Each cylinder must be subjected to a process treatment after welding and before stress relieving by hydrostatically pressurizing to at least 100 percent but not more than 110 percent of the test pressure and maintained at this pressure for 3 minutes. The process treatment must be witnessed by the independent inspector.

§ 178.44(e) *Welding or Brazing*.

- (1) Welding as prescribed in § 178.44(a) of this special permit is authorized.
- (2) All pressure welds must be 100 percent radiographed after hydrostatic test.

§ 178.44(f) *Wall Thickness*.

- (1) Applies except that 120,000 psi replaces 105,000 psi in the last sentence.
- (2) and (3) * * *

§ 178.44(g) *Heat Treatment.*

The half cylinders may be stress relieved or annealed for forming. Welded cylinders must be stress relieved at a temperature of $900^{\circ}\text{F} \pm 25^{\circ}$ for one hour. Stress relieving to be done after process treatment, and before hydrostatic test.

§ 178.44(j) *Cycling Test.*

* * *

(1) * * *

(2) * * *

(3) A “lot” is defined as a group of cylinders sequentially fabricated conforming to a specific design, from the same heat of steel, manufactured by the same process, and stress relieved under the same conditions of time, temperature and atmosphere and may not exceed 200 cylinders.

(4) * * *

(5) Cycle test on each lot prescribed in paragraph (3) above is not required when the 3 cylinders tested to qualify the design meet the requirements of paragraph § 178.44(p)(4)(i) of this special permit. The results of design qualification testing per § 178.44(j) and § 178.44 (p)(4) of this special permit must be submitted to the AAHMS prior to initial shipment.

§ 178.44(m) *Physical tests*

Not required.

§ 178.44(n) *Magnetic particle inspection.*

Not required. Instead each cylinder must be inspected using apparatus and procedures for liquid penetrant examination in accordance with ASTM-E-165-65. Inspection must be performed externally on the finished container after hydrostatic test. Evidence of discontinuities, which in the opinion of the independent inspector may appreciably weaken or decrease the durability of the cylinder, must be cause for rejection.

§ 178.44(p) *Acceptable results of tests.*

- (1) Flattening required without cracking to six times the wall thickness of the cylinder.
- (2) Physical tests - not required.
- (3) Burst pressure must be at least $\frac{4}{3}$ times the test pressure. Failure must be in the sidewall in a longitudinal direction. Actual burst pressure must be recorded.
- (4) Cycling -
 - (i) Design qualification test - At least 50,000 cycles from 0 to service pressure without evidence of distortion or failure.
 - (ii) Lot qualification test - At least 10,000 cycles from zero to service without evidence of distortion or failure.

§ 178.44(q) *Rejected cylinders.*

Repair of welded seams by welding prior to process treatment is authorized; subsequent thereto containers must be heat treated and pass all prescribed tests.

§ 178.44(r) *Marking.*

Applies except that the cylinders must be marked "DOT-SP 8220" instead of "DOT-3HT" followed by the service pressure and stamping elastic expansion is not required.

§ 178.44(s) *Inspectors Report.*

Inspectors report must be appropriately modified to reflect identification and compliance with this special permit.

- b. TESTING: Each cylinder must be reinspected and hydrostatically retested every 3 years in accordance with § 180.205 as prescribed for DOT-3HT cylinders, except that the rejection elastic expansion criteria does not apply.

8. SPECIAL PROVISIONS:

- a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

- b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modifications or changes are made to the package and it is offered for transportation in conformance with this special permit and the HMR.
 - c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
 - d. Each packaging manufactured under the authority of this special permit must be marked with a registration symbol designated by the OHMS for a specific manufacturing facility.
 - e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.
 - f. Cylinders are subject to the use restrictions prescribed in § 173.302a(a)(2), except they may also be used in military weapons systems.
 - g. Cylinders must be shipped in strong outside packagings in conformance with § 173.301(a)(9).
 - h. Cylinder authorized under this special permit are not authorized 15 years after the manufacture.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo aircraft only.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- 'The Hazardous Materials Safety and Security Reauthorization Act of 2005' (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: KH